

BOUNDNESS OF PAIRS WITH LOG GENERAL TYPE

1. INTRODUCTION

Let X be a smooth projective variety of general type, then by definition there exists an integer N such that the global sections of $\omega_X^{\otimes N}$ define a birational map to projective space

$$\phi_N : X \dashrightarrow \mathbb{P}H^0(\omega_X^{\otimes N}).$$

It is classically known that if $\dim X = 1$, then ϕ_3 is always birational. By a result of Bombieri, if $\dim X = 2$, then ϕ_5 is birational. More recently Tsuji, Hacon-McKernan [HM06] and Takayama [Ta06] proved that: *For any $n > 0$ there exists an integer $N = N(n) > 0$ such that ϕ_N is birational for any manifold of general type and dimension n .* The existence of such a bound is an important ingredient in constructing moduli spaces for varieties of general type. One should note however that (canonical models of) varieties naturally degenerate to “stable varieties” (this is already evident in the 1-dimensional case). It is therefore important to understand the geometry of these varieties. Restricting to an irreducible component, one reduces this problem to the study of the geometry of dlt pairs of general type with reduced boundary. In this paper we show the following natural generalization of the above mentioned result of Tsuji, Hacon-McKernan and Takayama.

1.1. Theorem. *Let $n \in \mathbb{Z}_{>0}$, then there exist $\delta \in \mathbb{R}_{>0}$ and $N \in \mathbb{Z}_{>0}$ such that if $(X, D = \sum_{j \in I} D_j)$ is an n -dimensional dlt pair with reduced boundary such that $K_X + D$ is big, then*

- (i) $\text{vol}(K_X + D) \geq \delta$; and
- (ii) *there is an integer N such that $|N(K_X + D)|$ is birational.*

When X is not uniruled we have the following more precise boundedness result.

1.2. Theorem. *Let $n \in \mathbb{Z}_{>0}$ and $\epsilon \in \mathbb{R}_{>0}$. Then there exist $N, M \in \mathbb{Z}_{>0}$ and $\delta \in \mathbb{R}_{>0}$ depending only on n and ϵ such that if X is a nonuniruled variety, $\dim X = n$, $(X, D = \sum d_i D_i)$ a dlt pair with $d_i \geq \epsilon$ and $K_X + D$ is big, then the following are true*

- (i) $\text{vol}(K_X + D) \geq \delta = \delta(n)$;
- (ii) $|N(K_X + D)|$ is birational; and
- (iii) *if $K_X + D$ is ample, then $|M(K_X + D)|$ is very ample.*

Next we investigate the properties of pluricanonical maps of klt varieties of general type. It is hoped that one may construct reasonable moduli spaces for these varieties. Once again, boundedness properties are an essential ingredient of these constructions. The natural conjecture in this context is the following.

1.3. Conjecture. *Let $\epsilon \in \mathbb{R}_{>0}$, $n \in \mathbb{Z}_{>0}$ and $\mathcal{A}$ be a dcc set then there exists a constant $\delta = \delta(\epsilon, n, \mathcal{A})$ (only depending on $\epsilon, n, \mathcal{A}$) such that for any n -dimensional ϵ -lc pair $(X, D = \sum_{j \in I} d_j D_j)$ with coefficients $d_i \in \mathcal{A}$, we have that*

$$\text{vol}(K_X + D) \geq \delta.$$

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One should also note that this conjecture is one of the main steps necessary to prove the conjecture on effective litaka fibration for pairs (Cf. [TX08], Conjecture 1.1). We prove the following two results

1.4. Theorem. *Conjecture 1.3 holds if $\dim X \leq 3$, or*

1.5. Theorem. *Let $n \in \mathbb{Z}_{>0}$ and $\epsilon, \delta \in \mathbb{R}_{>0}$, then there exist integers $N, M \geq 0$ such that if $(X, \Delta = \sum d_i D_i)$ is a non-uniruled n -dimensional ϵ -lc pair where $d_i \geq \delta$, then*

- (1) *if $K_X + D$ is big, then $|N(K_X + D)|$ is birational, and*
- (2) *if $K_X + D$ is ample, then $M(K_X + D)$ is very ample.*

The main idea of the proof of (1.1) is similar the one in [HM06] and [Ta06]: first we show that if there are some constants a_n and b_n which only depends on the dimension X , such that the linear system

$$(1) \quad \left| \left[a_n + \frac{1}{\text{vol}(K_X + D)} b_n \right] (K_X + D) \right|$$

gives a birational map. Then show that if $\text{vol}(K_X + D) \leq 1$, then all such pairs (X, D) belong to a birationally bounded family of pairs. In order to establish the first claim, one uses the sections of $K_X + D$ to produce a log canonical center V of $(X, D + \mathcal{D})$ passing through a very general point $x \in X$ where $\mathcal{D} \sim_{\mathbb{Q}} k(K_X + D)$ and $k = O(\text{vol}(K_X + D)^{-1/n})$. We must then “cut down” V until it has dimension 0. To achieve this, it is necessary to bound the volume of $(K_X + D + \mathcal{D})|_V$. This is done by using Kawamata’s subadjunction formula and the key observation is that if the open variety $U = X - \text{Supp}(D)$ is of log general type and so is any subvariety $V \subset U$ passing through a very general point $x \in U$.

2. PRELIMINARIES

In this section, we introduce some tools which we will need in our argument.

2.1. Open varieties. In this section we will recall some result on the birational geometry of open varieties. For a more detailed account, we refer the reader to [Ii81].

¹I added ϵ -lc.

2.1. Definition. Let U be a smooth quasi-projective variety. A *log smooth compactification* of U is a smooth projective variety X and a reduced simple normal crossing divisor D on X such U is isomorphic to $D := X \setminus U$ is. We define the *log Kodaira dimension* $\kappa(U)$ of U to be the Iitaka dimension $\kappa(X, K_X + D)$ of the adjoint bundle $K_X + D$. Note that log smooth compactifications always exist.

It is not hard to show that $\kappa(U)$ satisfies the following properties (cf. [Ii81], Lecture 6),

- (1) it does not depend on the choice of log smooth compactification X ;
- (2) if $U_1 \rightarrow U_2$ is a birational proper morphism, then $\kappa(U_1) = \kappa(U_2)$.

2.2. Definition. We say U is of *log general type* if

$$\kappa(U) = \dim(U).$$

The following lemma will be important is well-known, but we could not find a reference in the literature.

2.3. Lemma. *Let U be a smooth quasi-projective variety of log general type and V be a subvariety passing through a very general point $x \in U$ then V is of log general type i.e. any desingularization V^{sm} of V is of log general type.*

Proof. Let X be a log smooth compactification of U . If the statement is not true, then by a Hilbert scheme argument, we know that there is a subscheme $B \subset \text{Hilb}(X)$ with a dominating universal family $f : Y/B \rightarrow X$, such that for the generic fiber Y_η , the desingularization of $Y_\eta \setminus f^{-1}(D)$ is not of log general type.

$$\begin{array}{ccc} Y_U & \xrightarrow{\quad} & U \\ \downarrow & \square & \downarrow \\ Y & \xrightarrow{f} & X \\ \downarrow & & \\ B & & \end{array}$$

By cutting B with general hypersurfaces, we can assume f is a generically surjective morphism. Then after desingularization, we can assume that $(Y, f^{-1}(D))$ is a log smooth pair. Thus

$$\dim(Y_U) \geq \kappa(Y_U) \geq \kappa(U) = \dim(U),$$

and we conclude that $\dim(Y_U) = \kappa(Y_U)$. However, from ([Ii81], Lecture 8, Theorem 2), we obtain

$$\kappa(Y_U) \leq \kappa(Y_U)_\eta + \dim(B) < \dim(Y_\eta) + \dim(B) = \dim(Y),$$

which is absurd. □

2.2. Kawamata Subadjunction. This was first established in [Ka98], see also [Ko07].

2.4. Definition. [Ko07] (*Standard normal crossing assumptions*) We say that a projective fibration $f : S \rightarrow V$ together with $\mathbb{Q}$ -divisors R, B satisfies the standard normal crossing assumptions if the following hold

- (1) S, V are smooth,
- (2) $R + f^*B$ and B are *snc* divisors,
- (3) f is smooth over $V \setminus B$, and
- (4) R is a relative *snc* divisor over $V \setminus B$.

2.5. Remark. Replacing S and V by appropriate birational models, we can always assume that the standard normal crossing assumptions are satisfied.

2.6. Theorem (Kawamata's canonical bundle formula). *Let $f : S \rightarrow V$ and R, B satisfy the standard normal crossing assumptions. Let F be the generic fibre of f . Assume that $K_S + R \sim_{\mathbb{Q}} f^*H_R$ for some $\mathbb{Q}$ -divisor H_R on V . Let $R = R_h + R_v$ be the horizontal and vertical parts of R and assume that when we write $R_h = R_{h(\geq 0)} - R_{h(\leq 0)}$ as the difference of its positive and negative parts, we have $h^0(F, [R_{h(\leq 0)}]_F) = 1$. Then we can write*

$$K_S + R \sim_{\mathbb{Q}} f^*(K_V + J(S/V, R) + B_R)$$

where

- (1) The moduli part $J(S/V, R)$ is nef and depends only on $(F, R_h|_F)$ and V .
- (2) The boundary part B_R depends only on $f : S \rightarrow V$ and R_v . More precisely, B_R is the unique smallest $\mathbb{Q}$ -divisor supported on B such that

$$\text{red}(f^*B) \geq R_v + f^*(B - B_R).$$

- (3) The pair (V, B_R) is lc iff The pair (S, R) is lc.
- (4) B_i appears in B_R with nonnegative coefficient if $f_*\mathcal{O}_S([-R_v]) \subset \mathcal{O}_{B_i, V}$.

Proof. See [Ko07], Theorem 8.5.1. □

2.7. Remark. In the above formula, $J(S/V, R)$ is only defined as a $\mathbb{Q}$ -divisor class. However, B_R is a $\mathbb{Q}$ -divisor.

2.3. Dlt modification. The following technical result of the existence of *dlt* modifications turns out to be very useful in the study of the geometry of log canonical centers.

2.8. Proposition (Hacon). *Let X be a normal quasi-projective variety and B a boundary $\mathbb{R}$ -divisor on X such that $K_X + B$ is $\mathbb{R}$ -Cartier. There exists a projective birational morphism $f : Y \rightarrow X$ from a normal quasi-projective variety such that*

- (1) Y is $\mathbb{Q}$ -factorial;
- (2) $a(E, X, B) \leq -1$ for every f -exceptional divisor E on Y ;

(3) if $B_Y = f_*^{-1}(B) + \sum_{E \in \text{Ex}(f)} E$, then (Y, B_Y) is dlt and

$$K_Y + B_Y = f^*(K_X + B) + \sum_{a(E, X, B) < -1} (a(E, X, B) + 1)E.$$

Moreover, for any finite connection of divisors $\{E_i\}$ with $a(E_i, X, B) = -1$ and the center of E_i on X is of codimension at least 2, we can assume that

$$\{E_i\}_i = \{E \mid E \text{ is } f\text{-exceptional and } a(E, X, B) = 1\},$$

i.e., the set of divisors extracted by f with log discrepancy 0 is precisely $\{E_i\}$.

Proof. A proof will be added. $\square$

3. LIFTING LOG CANONICAL CENTERS

In this section, we are going to show that

3.1. Theorem. *Let (X, D) be a dlt pair with reduced boundary such that $K_X + D$ is big. Then there exists constants a_n and b_n which only depend on the dimension $n = \dim X$, such that*

$$\lceil a_n + \frac{b_n}{\text{vol}(K_X + D)^{1/n}} \rceil (K_X + D)$$

is birational.

Let $K_X + D \sim_{\mathbb{Q}} A_X + B_X$ where A_X is a general ample $\mathbb{Q}$ -divisor and $B_X \geq 0$. For $t \gg 0$, $K_X + D + tA_X$ is ample and by [BCHM06] (since $D + A_X \sim_{\mathbb{Q}} D'$ where (X, D') is klt and D' is big) we may run the $K_X + D + A_X$ mmp with scaling of A_X . We may therefore assume that $K_X + D + A_X$ is nef.

Let $x \in X$ be a general point and let

$$\mathcal{D} \sim_{\mathbb{Q}} k_0(K_X + D)$$

such that $(X, D + \mathcal{D})$ is lc but not klt near x . We may assume that $k_0 = O(\text{vol}(K_X + D)^{-1/n})$ and that $\text{LCC}(x, X, D + \mathcal{D}) = \{V\}$ is an irreducible subvariety, which is dominated by a unique exceptional divisor S of log discrepancy 0.

Let $f : Y \rightarrow (X, D + \mathcal{D})$ be a dlt modification which extracts S as in (2.8). We write

$$(D + \mathcal{D})_Y = S + R \quad \text{and} \quad R = R_0 + R_1,$$

where R_0 is the sum of those components with coefficient 1.

We now choose appropriate resolutions $r_V : V' \rightarrow V$, $r_Y : Y' \rightarrow Y$ and $r_S : S' \rightarrow S$ such that we have the following diagram

$$\begin{array}{ccccc}
& & S' & \xrightarrow{i_{S'}} & Y' \\
& \nearrow \gamma & \downarrow r_S & & \downarrow r_Y \\
V' & & S & \xrightarrow{i_S} & Y \\
& \searrow \gamma_V & \downarrow \pi & & \downarrow f \\
& & V & \xrightarrow{i_V} & X
\end{array}$$

If we write

$$\Gamma + (f \circ r_Y)^*(K_X + D + \mathcal{D}) = K_{Y'} + S' + R',$$

such that Γ, S' and R' are effective divisors without common components, then we can assume that

$$\pi' : S' \rightarrow V' \quad \text{and} \quad \text{Supp}((\Gamma + R')|_{S'}), \quad \text{Supp}(r_V^{-1}(D \cup \mathcal{D})),$$

satisfy the standard normal crossing assumptions as in (2.4).

We now apply (2.6) to obtain the following comparison of boundary divisors, which will be needed in the induction process.

3.2. Proposition. *If we write*

$$(i_V \circ r_V)^*(K_X + D + \mathcal{D}) = K_{V'} + J(S'/V', (R' - \Gamma)|_{S'}) + B,$$

as in (2.6), where

- (1) B is the boundary part which is $\mathbb{Q}$ -divisor, and
- (2) $J = J(S'/V', (R' - \Gamma)|_{S'}) +$ is the moduli part, which is a nef $\mathbb{Q}$ -line bundle,

then

- (1) $K_V + \text{red}(\pi_*(S \cap R_0))$ is a big Weil divisor;
- (2) $\text{red}(\pi_*(S \cap R_0)) \leq (r_V)_*(B)$.

Proof. (1): Since x is a very general point on $X \setminus D$, any desingularization V^{sm} of $V \setminus (V \cap D)$ is of log general type (2.3). Therefore $K_V + \text{red}(V \cap D)$ is a big Weil divisor. So it suffices to show that

$$\text{red}(V \cap D) \subset \pi_*(S \cap R_0).$$

We observe that for any $v \in V \cap D$, $f^{-1}(v)$ is connected and intersects with S and R_0 . Moreover every exceptional divisor of f is contained in R_0 , and the statement follows easily.

(2): By Kawamata subadjunction cf. (2.6.2), for any divisor B_i in B , its coefficients is not less than 1, if (and only if) it is dominated by a log canonical

center of $(S', (R' - \Gamma)|_{S'})$. If $B_i \subset R_0 \cap S$, $(r_Y)_*^{-1}(R_0)$ may not intersect S' . Nevertheless, from the Kollár-Shokurov connectedness theorem ([KM98], Theorem 5.48), we know that over a general point η_{B_i} of B_i , the intersection of $f \circ r_Y^{-1}(\eta)$ with the components of $(S' + R' - \Gamma)$ whose coefficients are larger or equal to 1 is connected. In particular, there is a component of coefficient at least one in $R'|_{S'} \subset S'$ which dominates η_{B_i} . $\square$

By induction, we know that the log volume of V^{sm} is bounded below. We define

$$\overline{B} = \sum_{B_i \subset (i_V \circ r_V)^{-1} \text{Supp}(D)} \text{red}(B_i) \quad \text{and} \quad \overline{R}_{S'} = R^{\text{hor}}|_{S'} + \sum_{R_i \subset (i_V \circ \pi \circ r_S)^{-1} \text{Supp}(D)} \text{red}(R_i)|_{S'}$$

where R^{hor} consists of those components of R whose intersection with S' dominates V' and the R_i are the components of $\text{Supp}(R')$.

3.3. Proposition. *Let H be any ample $\mathbb{Q}$ -divisor on Y' . For any sufficiently divisible integer $d > 0$, there is an injection*

$$|d\pi'^*(K_{V'} + \overline{B})| \hookrightarrow |d(K_{S'} + \overline{R}_{S'} + H)|_{S'},$$

induced by adding a divisor $d\Psi \geq 0$, with the property that after restricting on a general fiber of $S' \rightarrow V'$,

$$d\Psi|_{S'_\eta} \leq d\Gamma|_{S'_\eta}.$$

Proof. From the definition,

$$\pi'^*(K_{V'} + B + J) = K_{S'} + R'|_{S'} - \Gamma|_{S'}.$$

If $B = \sum a_i B_i$ and $(R' - \Gamma)|_{S'} = \sum r_j R_j$, then by definition (cf. (2.6)), a_i is the smallest number such that

$$n_j(1 - a_i) \leq 1 - r_j,$$

where r_j runs over the coefficients of all components R_j of $(R' - \Gamma)|_{S'}$ which are mapped onto B_i , and n_j is the corresponding ramification index. It follows that $n_j a_i \geq r_j + n_j - 1 \geq r_j$, for each R_j mapping onto B_i . Therefore, $a_i < 0$ implies that $r_j < 0$ for all j . We thus have

$$\pi'^*\left(\sum_{B_i \not\subset (i_V \circ r_V)^{-1}(D), a_i < 0} a_i B_i\right) \geq \sum_{r_j < 0} r_j R_j|_{S'}.$$

Therefore

$$\begin{aligned} K_{V'} + \overline{B} + J + \sum_{B_i \not\subset (i_V \circ r_V)^{-1} \text{Supp}(D), a_i > 0} a_i B_i = \\ (K_{V'} + B + J) - \sum_{B_i \subset (i_V \circ r_V)^{-1} \text{Supp}(D)} (a_i - 1) B_i - \sum_{B_i \not\subset (i_V \circ r_V)^{-1} \text{Supp}(D), a_i < 0} a_i B_i. \end{aligned}$$

But from our discussion above, the pull back of this term by π' is less or equal to

$$K_{S'} + (R' - \Gamma)|_{S'} - \sum_{R_j \subset (i_V \circ \pi \circ r_S)^{-1} \text{Supp}(D)} (r_j - 1) R_j - (-\Gamma|_{S'}),$$

which is $K_{S'} + \overline{R}_{S'}$.²

Finally, since J is nef $J + H|_{S'}$ is $\mathbb{Q}$ -effective. So there is an effective divisor Ψ , inducing the injective homomorphism in the statement. It is easy to see that Ψ satisfies the required inequality. $\square$

We can extend the sections from S to Y , more precisely,

3.4. Proposition. *Let A_Y be a very ample divisor of general position on Y , such that $K_Y + S + R + A_Y$ is very ample and Cartier. For any $n > 0$ sufficiently divisible we define the Cartier divisor*

$$L_n = f^*(n(K_X + D + (k_0 + 1)A_X + \mathcal{D})) + K_Y + S + R + A_Y.$$

Then we have a surjective homomorphism

$$H^0(Y, \mathcal{O}_Y(L_n)) \twoheadrightarrow H^0(S, \mathcal{O}_S(L_n|_S)).$$

Proof. Since $(Y, S + R)$ is dlt and $L_n - (K_Y + S + R)$ is ample, it follows from Kawamata-Viehweg vanishing that $H^1(Y, \mathcal{O}_Y(L_n - S)) = 0$. $\square$

3.5. Lemma. *There exists a rational number $\delta > 0$ which only depends on the dimension of V' , such that if $v \in V'$ is a very general point, then there exists a $\mathbb{Q}$ -divisor $\Theta_{V'} \sim_{\mathbb{Q}} k(K_{V'} + \overline{B})$ such that a general point $v \in V'$ is a log canonical center of $K_{V'} + \Theta_{V'}$.*

Proof. By our choice of x , any desingularization of V^0 is of log general type. This implies that $K_{V'} + \overline{B}$ is big and so the statement follows by induction on the dimension. $\square$

As $v \in V'$ is very general, we can assume that r_V is isomorphism around v .

3.6. Lemma. *We may find a positive integer n_0 and a positive integer λ , with the following properties: For every sufficiently large and divisible integer n , there is an effective divisor G_n on Y such that*

- (1) G_n does not contain S ,
- (2) $G_n + \lambda S \sim_{\mathbb{Q}} f^*((n_0 + n(3 + 2k_0))(K_X + D))$,
- (3) $G_n|_S = (r_S)_*(\pi')^*(\frac{n}{k}\Theta_{V'}) + B_n$, for some effective B_n , and
- (4) $B_n|_{S_\eta}$ belongs to a fixed linear system when restricting on a general fiber.

Proof. By what we have seen above, we have that

$$\frac{d}{k}\pi'^*\Theta_{V'} + d\Psi \in |d(K_{S'} + \overline{R}_{S'} + H|_{S'})|.$$

We have that $f^*(K_X + D + \mathcal{D}) - (K_Y + S + R) \geq 0$. The different (cf. [Ko92]) $\text{Diff}(S, Y, R)$ is an effective divisor such that

$$(K_Y + S + R)|_S = K_S + \text{Diff}(S, Y, R).$$

²actually it is $K_{S'} + \overline{R}_{S'} \vee R_{S'}^h$!

Since Y is a dlt model, R is a reduced effective divisor whose support equals the exceptional locus. Therefore as $\text{Diff}(S, Y, R) = \text{Diff}(S, Y) + R|_S$, we have that for each prime divisor P contained in $(i_S \circ f)^{-1}\text{Supp}(D)$, its coefficient in $R|_S$ is at least one. Since $K_S + \text{Diff}(S, Y, R)$ is dlt, we conclude that the coefficient of P in $K_S + \text{Diff}(S, Y, R)$ is one. Therefore $(r_S)_* \overline{R}_{S'} \leq \text{Diff}(S, Y, R)$. Notice that we also have that

$$f^*(K_X + D + \mathcal{D})|_S - (r_S)_*(K_{S'} + \overline{R}_{S'}) = \Xi_1 \geq 0$$

is effective. But its restriction to a general fiber of π is 0 and hence its support is contained in $\pi^{-1}(Z)$ for some proper closed subset $Z \subset V$.

So we obtain from (3.4), that the image of

$$(r_S)_*|n(K_{S'} + \overline{R}_{S'} + H|_{S'})| \hookrightarrow |n(K_S + \text{Diff}(S, Y, R) + H|_S)| \\ \hookrightarrow |(nf^*(K_X + D + (k_0 + 1)A_X + \mathcal{D} + H) + K_Y + S + R + A_Y)|_S|$$

(by adding a general divisor Ξ_0 in $|K_Y + S + R + A_Y|$) can be lifted to elements of

$$|f^*n(K_X + D + (k_0 + 1)A_X + \mathcal{D} + H) + (K_Y + S + R + A)|.$$

From (3.3), denote by S_n the lift of

$$n(r_S)_*(\pi'^* \frac{1}{k} \Theta_{V'} + \Psi) + n(\Xi_1 + f^*((k_0 + 1)A_X + H)|_S) + \Xi_0$$

to

$$|nf^*(K_X + D + (k_0 + 1)A_X + \mathcal{D} + H) + (K_Y + S + R + A_Y)|.$$

We can fix a positive integer n_0 , and an effective divisor

$$S_0 \in |n_0 f^*(K_X + D) - (K_Y + R + S + A_Y)|.$$

Then

$$S_n + S_0 + n(k_0 + 1)f^*B_X \in |(n_0 + n(3 + 2k_0))f^*(K_X + D)|.$$

Let G_n satisfy that

$$G_n + \lambda S = S_n + S_0,$$

where G_n does not contain S . Since S_n does not contain S and S_0 is fixed, λ is fixed. It follows that we can write

$$G_n|_S = (r_S)_*(\pi')^*(n\Theta_{V'}) + B_n.$$

When we restrict to the general fiber S_η , we have

$$B_n - (r_S)_*(\pi')^*(\frac{n}{k}\Theta_{V'}) - S_0 = \Lambda_n \geq 0$$

and there is a open subset $U \subset V$ which does not depend on v and n , such that for any $\eta \in U$,

$$\Lambda_n|_{S_\eta} = (\Xi_0 + n(r_S)_*\Psi)|_{S_\eta} \leq \Xi_0|_{S_\eta}$$

as $\Psi|_{S'_\eta} \leq \Gamma|_{S'_\eta}$ and $(r_S)_*\Gamma|_{S_\eta} = 0$. □

Let G_n be the sequence of divisors we constructed above. Set

$$\Gamma_n = (G_n + \lambda S)/n, \quad \Theta_n = f_*(\Gamma_n) \quad \text{and} \quad \Delta_n = \alpha_n \mathcal{D} + \Theta_n.$$

Where $\alpha_n \leq 1$, such that the log discrepancy of S with respect to $K_X + \Delta_n$ is zero. And when n approaches infinity, α_n approaches 1. And now we have

3.7. Proposition. *W is a pure log canonical center of (X, Δ_n) .*

Proof. Since this result is local about the generic point of V , we may pass to the open set U defined as above. In this case

$$K_Y + S + F_n + G_n/n = f^*(K_X + \Delta_n), \text{ where}$$

$$K_Y + c_n S + F_n = \pi^*(K_X + \alpha_n \Delta),$$

F_n is a divisor, which does not contain S and $c_n = 1 - \lambda/n < 1$ is a positive number. Restricting to S , we get

$$K_S + F_n|_S + \pi^*(\Theta_{V'}) + B_n/n.$$

Let Z be the inverse image of W . $V' \rightarrow V$ is a log resolution such that $E \subset V'$ is a divisor centered at W with log discrepancy 0.

$$\begin{array}{ccc} S' & \xrightarrow{r_S} & Z \subset S \\ \pi' \downarrow & & \downarrow \pi \\ E \subset V' & \xrightarrow{r_V} & W \subset V \end{array}$$

For n big enough, after possible further shrinking U , we can assume $K_S + F_n|_S + B_n/n$ is *klt* since (S, F_n) is *klt*, and $B_n|_{S_\eta} = S_0|_{S_\eta}$, where S_0 is a fixed divisor.

Since we can assume that $S' \rightarrow V'$ is smooth over $\Theta_{V'}$, then $K_Y + S + F_n + G_n/n$ has a pure log canonical center along Z , because $(r_S)^*(K_Y + S + F_n + G_n/n)$ has a pure log canonical center along $(\pi')^{-1}E$. Then we can apply inversion of adjunction to obtain the conclusion. $\square$

Proof of Theorem (3.1). By what we have seen above (cf. (3.6) and (3.7)), there exists a divisor $\Delta_n \sim_{\mathbb{Q}} \beta(K_X + D)$ where $\beta = O(\text{vol}(K_X + D)^{-1/\dim X})$ such that a general point of X is a component of the cossupport of $\mathcal{J}(\Delta_n)$. The result now follows easily as in [HM06] and [Ta06]. $\square$

4. BOUNDESS OF THE BOUNDARY

Let (X, D) be a log smooth pair of dimension n with reduced boundary. Let $m = a_n + \frac{b_n}{\text{vol}(K_X + D)^{\frac{1}{n}}}$, which we get in the previous section.

4.1. Theorem. $f : X \rightarrow Y$ is a birational morphism such that $m(K_X + D) \sim f^*L + E$, where $L \sim (n^2 + n + 2)H$, where H is a very ample divisor on Y and E effective. If $\text{vol}(K_X + D)$ is bounded, then

$$\deg(f_*(E)) = f^*(L)^{n-1} \cdot E$$

is bounded.

Proof. After further blow up, we can assume $(X, \text{red}(D + E))$ is log smooth. A reduced component F_i of E is not contracted by f if and only if $f^*(H)|_{F_i}$ is big and nef. Let

$$F = \sum n_i F_i, \text{ where } F_i \text{ is not contracted by } f \text{ and } m_i = \text{mul}_{E_i} E.$$

Then we have $\deg f_* E = \deg f_* F$.

4.2. Lemma. $K_{\text{red}(F)} + \frac{n^2+n+2}{2}H|_{\text{red}(F)}$ separates the points near which f induces an isomorphism on $\text{red}(F)$.

Proof. Applying the argument in [AS95] (see also [Ko97], Section 5 for an algebraic treatment) To be added????????? $\square$

Then we see

$$K_X + (n^2 + n + 2)/2 f^* H \quad \text{and} \quad K_{\text{red}(F)} + (n^2 + n + 2)/2 f^* H|_{\text{red}(F)}$$

give birational maps.

We define $K_X = (K_X + F)|_F$.

4.3. Lemma. There is an injective that $K_{\text{red}(F)} \hookrightarrow K_F$.

Proof. Let $F = \sum n_i F_i$ ($n_i \geq 1$). Tensoring

$$0 \rightarrow \mathcal{O}_X(-\text{red}(F)) \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_{\text{red}(F)} \rightarrow 0,$$

by $\mathcal{O}_X(\sum(-n_i + 1)F_i)$, we get

$$0 \rightarrow \mathcal{O}_X(\sum -n_i F_i) \rightarrow \mathcal{O}_X(\sum(-n_i + 1)F_i) \rightarrow \mathcal{O}_{\text{red}(F)}((-n_i + 1)F) \rightarrow 0.$$

Hence

$$0 \rightarrow \mathcal{O}_{\text{red}F}(\sum(-n_i + 1)F_i) \rightarrow \mathcal{O}_F \rightarrow \mathcal{O}_{\sum(n_i-1)F_i} \rightarrow 0.$$

Tensoring with $K_X + F$, we obtain the claim. $\square$

From the above argument, we conclude that $\mathcal{O}(K_F + 1/2 f^* L)$ has a nontrivial global section with nonzero restriction s_F on each component F_i of F . Furthermore, this section can be lifted to a global section of $\mathcal{O}(K_X + F + 1/2 f^* L)$. We

fix a section $s \in H^0((K_X + F + 1/2f^*L)$, Now look at the following commutative diagram

$$\begin{array}{ccccccc}
0 & \longrightarrow & \mathcal{O}(K_X + d/2f^*L) & \longrightarrow & \mathcal{O}(K_X + F + d/2f^*L) & \longrightarrow & \mathcal{O}_F(K_F + d/2f^*L) \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & \mathcal{O}(dK_X + (d-1)F + df^*L) & \longrightarrow & \mathcal{O}(dK_X + dF + df^*L) & \longrightarrow & \mathcal{O}_F(dK_F + df^*L) \longrightarrow 0
\end{array}$$

The vertical morphisms are induced by adding $\frac{d-1}{2}s + l$, where l is a general section of $f^*(\frac{n^2+n+2}{2}H)$. Let d be an even number, since $H^1(X, K_X + d/2f^*L) = 0$, we obtain

$$H^0(K_X + F + df^*(L)) \twoheadrightarrow H^0(K_F + df^*L|_F).$$

In particular, we have

$$h^0(d(K_X + F + f^*L)) \geq h^0(dK_F + df^*L) + h^0(dK_X + (d-1)F + df^*L).$$

Since $H^0(K_X + f^*L) \neq 0$, we conclude that

$$h^0(dK_X + (d-1)F + df^*L) \geq h^0((d-1)(K_X + F + f^*L)).$$

From

$$h^0(dK_F + df^*L|_F) \geq h^0(d/2(L)|_F) = \frac{d^{n-1}}{2^{n-1}(n-1)!} \deg f_*(F),$$

we know there is a comparison of the volume that

$$\text{vol}(K_X + F + f^*L) \geq O(\deg f_*(F)).$$

from which we conclude that $\deg f_*(F)$ is bounded. $\square$

Then we get $\deg(f_*(D)) \leq \deg \frac{1}{m}(L + f_*E) - f_*(K_X) = \deg \frac{1}{m}(L + f_*E) - K_Y$, which is bounded.

5. BOUNDNESS OF THE BOUNDARY

In this section we will prove Theorem (1.1). Since (ii) follows from (i), by (3.1) it suffices to show that there exists $\delta = \delta(n) > 0$ such that $\text{vol}(K_X + D) \geq \delta$. We may assume that $\text{vol}(K_X + D) < 1$. Let

$$t = \lceil a_n + \frac{b_n}{\text{vol}(K_X + D)^{\frac{1}{n}}} \rceil.$$

5.1. Lemma. *We may assume (after possibly rechoosing t) that $K_X + t(K_X + D)$ induces a birational map $f : X \dashrightarrow X^b$ and that $\frac{t+1}{2}K_X + \frac{t-1}{2}D$ is linearly equivalent to an effective divisor.*

Proof. This follows easily from (3.1). $\square$

From [BCHM06], we know $(X, \frac{t}{t+1}D)$ has a minimal model $g : (X, D) \dashrightarrow (X^m, D^m)$, hence

$$\text{vol}(K_X + \frac{t}{t+1}D) = (K_{X^m} + \frac{t}{t+1}D^m)^n.$$

We now choose a resolution $\pi : \overline{X} \rightarrow X$ which resolves the undetermined loci of f and g

$$\begin{array}{ccc} \overline{X} & \xrightarrow{\overline{f}} & X^b \\ \downarrow \overline{g} & \nearrow \nu & \vdots f \\ X^m & \xleftarrow{g} & X \end{array}$$

5.2. Proposition. $h_*g_*(D)$ and $h_*g_*(K_X + D)$ have bounded degree.

Note that $h_*g_*(D)$ is not necessarily equal to $f_*(D)$ since h may extract some divisors.

Proof. We can write

$$\pi^*(K_X + t(K_X + D)) \sim_{\mathbb{Q}} L + F,$$

where L is base-point-free and F is the fixed part.

5.3. Lemma. $L \leq \overline{g}^*((t+1)K_{X^m} + tD^m)$ as $\mathbb{Q}$ -divisors.

Proof. Assume N is sufficiently divisible, then $N((t+1)K_{X^m} + tD^m)$ is base point free and as g is a minimal model of $(X, \frac{t}{t+1}D)$, we have that

$$|N\pi^*(K_X + t(K_X + D))| = |N\overline{g}^*((t+1)K_{X^m} + tD^m)| + F_N$$

and so $F_N \leq NF$ and the desired inequality follows. $\square$

5.4. Lemma. $\overline{g}^*((t+1)K_{X^m} + tD^m) \cdot L^{n-1} \leq \overline{g}^*((t+1)K_{X^m} + tD^m)^n$.

Proof. Since $\overline{g}^*((t+1)K_{X^m} + tD^m) \geq L$ are nef divisors, we have that

$$\overline{g}^*((t+1)K_{X^m} + tD^m)^i \cdot L^{n-i} \leq \overline{g}^*((t+1)K_{X^m} + tD^m)^{i+1} \cdot L^{n-i-1}$$

for $i = 1, 2, \dots, n-1$. The Lemma follows. $\square$

Since $\overline{f}^* \mathcal{O}_{X^b}(1) = L$, applying the projection formula, we have

$$\deg(h_*(D^m)) = \overline{g}_*^{-1}(D^m) \cdot L^{n-1} \leq \overline{g}^*(D^m) \cdot L^{n-1}.$$

Since $\frac{t+1}{2}K_X + \frac{t-1}{2}D$ is linearly equivalent to an effective divisor, so is $(t+1)K_X + (t-1)D$ and hence

$$\overline{g}^*(D^m) \cdot L^{n-1} \leq \overline{g}^*((t+1)K_{X^m} + tD^m) \cdot L^{n-1}.$$

Therefore, $h_*g_*(D) = h_*(D^m)$ has degree bounded by

$$(t+1)^n \text{vol}(K_X + \frac{t}{t+1}D) \leq (t+1)^n \text{vol}(K_X + D).$$

Since $t = O((\text{vol}(K_X + D))^{-1/n})$ and $\text{vol}(K_X + D) < 1$, the degree of $h_*g_*(D)$ is bounded. The claim for $h_*g_*(K_X + D)$ now follows easily. $\square$

5.5. Proposition. *There is a bounded family of pairs $(\mathcal{X}, \mathcal{D}) \rightarrow T$ parametrized by a scheme T of finite type, such that for any (X, D) with $\text{vol}(X, D) \leq 1$, there is a $t_0 \in T$ such that*

$$\text{vol}(X, D) \geq \text{vol}(\mathcal{X}_{t_0}, \mathcal{D}_{t_0}).$$

Proof. Let $D_b = h_*(D^m)$, whose supports form a bounded family by (5.2). Let $\mathcal{E}$ be the set of divisors in X^b which are exceptional over X^m . We claim that $\mathcal{E}$ also belongs to a bounded family. Since K_{X^b} and $h_*(K_{X^m})$ belong to bounded families, it suffices to show that the same holds for divisors $E \in \mathcal{E}$ with $a(E, X^m) = 0$.

Since the support of K_{X^b} Consider a log resolution $r : X' \rightarrow (X^b, D^b)$. Since the pairs (X^b, D^b) form a bounded family, we can choose X' in a bounded family $\mathcal{X}$. Consider now the divisors $D' = \text{Supp}(r_*^{-1}(D^b)) \cup \text{Ex}(r)$. We let $\mathcal{D}$ parametrize the divisors D' . Clearly $(\mathcal{X}, \mathcal{D})$ is again a bounded family. We denote the parametrizing space again by T . It suffices to prove the following.

5.6. Lemma. $\text{vol}(K_X + D) = \text{vol}(K_{X'} + D')$.

Proof. Let $\pi_1 : X'' \rightarrow X$ and $\pi_2 : X'' \rightarrow X'$ be a common resolution. Since

$$K_{X''} + (\pi_1^{-1})_*D + \text{Ex}(\pi_1) = \pi_1^*(K_X + D) + E,$$

where E is an effective exceptional divisor we have that

$$\text{vol}(K_{X''} + (\pi_1^{-1})_*D + \text{Ex}(\pi_1)) = \text{vol}(K_X + D).$$

On the other hand, since (X', D') is log smooth and D' is reduced with $D' = (\pi_1)_*((\pi_1^{-1})_*D + \text{Ex}(\pi_1))$ where the support of $(\pi_1^{-1})_*D + \text{Ex}(\pi_1)$ contains the support of $\text{Ex}(\pi_2)$, we have

$$K_{X''} + (\pi_1^{-1})_*D + \text{Ex}(\pi_1) = \pi_2^*(K_{X'} + D') + F,$$

where F is effective and exceptional so that

$$\text{vol}(K_{X''} + (\pi_1^{-1})_*D + \text{Ex}(\pi_1)) = \text{vol}(K_{X'} + D').$$

$\square$

$\square$

We end up the theorem with the following lemma, which should be well-known but we can not find in the literature.

5.7. Proposition. *If $\mathcal{X} \rightarrow T$ is a bounded family of varieties with a bounded family of line bundles $\mathcal{L}$, then the set*

$$V = \{v_t | v_t = \text{vol}(L_t), t \in T\} \setminus \{0\}$$

has a lower bound.

Proof. To be write if necessary. $\square$

5.8. Corollary. *For every integer $n > 0$ there exists a constant $\delta > 0$ and an integer $N > 0$ such that if (X, D) is a dlt pair with reduced boundary such that $K_X + D$ is big and $\dim X = n$, then*

- (1) $\text{vol}(K_X + D) \geq \delta$; and
- (2) $|N(K_X + D)|$ is birational.

Proof. Immediate from $\square$

5.9. Corollary. *Assume the existence of log canonical models for big dlt pairs.*

For every integers $n > 0$ and $M > 0$ there exists a an integer $N > 0$ such that if (X, D) is a lc pair with reduced boundary such that $K_X + D$ is ample, $\dim X = n$ and $(K_X + D)^n \leq M$, then $N(K_X + D)$ is very ample (and so (X, D) belongs to a bounded family).

Proof. To be added. $\square$

6. KLT CASE

6.1. Theorem. *Let $n \in \mathbb{Z}_{>0}$ and $\epsilon, \delta \in \mathbb{R} > 0$. Let $(X, D = \sum d_i D_i)$ be a klt pair such that $\dim X = n$, $d_i \geq \epsilon$ for all i and $K_X + (1 - \delta)D$ is pseudo-effective. Then $|N(K_X + D)|$ is birational.*

Proof. We may assume that (X, D) is log smooth. (add details!!!)

We then have that if $N\delta \geq 1/\epsilon$, then $(N+1)D \geq N(1-\delta)D + D_{\text{red}}$. The proof is now immediate from (1.1). $\square$

6.2. Corollary. *Let $n \in \mathbb{Z}_{>0}$ and $\epsilon \in \mathbb{R} > 0$. Then there exist integers $N, M > 0$ such that if $(X, D = \sum d_i D_i)$ is a klt pair, K_X is pseudo-effective, $\dim X = n$, $d_i \geq \epsilon$ for all i then*

- (1) *if $K_X + D$ is big then $|N(K_X + D)|$ is birational; and*
- (2) *if $K_X + D$ is ample, then $|M(K_X + D)|$ is very ample.*

6.3. Theorem. *Let $\epsilon \in \mathbb{R}_{>0}$ and let $\mathcal{A}$ be a dcc set. Let $(X, D = \sum d_i D_i)$ be a ϵ -lc³ 3-fold such that $d_i \in \mathcal{A}$ for all i and $K_X + D$ is big. Then $|N(K_X + D)|$ is birational.*

³do we really need the ϵ -lc condition? Should write non-bounded examples!

Proof. By (6.1), we may assume that $K_X + \frac{1}{2}D$ is not pseudo-effective. Let τ be the pseudo-effective threshold. In particular $0 \leq \kappa(K_X + \tau D) < 3$. We may assume that $K_X + D$ is nef and terminal, and we run a mmp with scaling of D , say $X_i \dashrightarrow X_{i+1}$, so that we have a $K_{X_N} + \tau D_N$ trivial Mori fiber space $X_N \rightarrow Z$. Note that X_N is terminal. If $\dim Z = 0$, then X_N is Fano and hence bounded. Arguing as above (???), it follows that (X, D) is birationally bounded.

If $\dim Z \in \{1, 2\}$, then the general fiber F of $X_N \rightarrow Z$ is terminal and Fano and hence bounded. But then τ belongs to a finite set (determined by $\mathcal{A}$; explain!!!) and so τD has coefficients in a dcc set. By the results on the effective Iitaka fibration (cf. [TX08]), there is an integer $R > 0$ such that $h^0(R(K_X + \tau D)) > 1$. Using Kollár's method we should now get the result. $\square$

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